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Machine learning as a feedback tool for the gestalt principle of proximity

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***Abstract*—When teaching students about the underlying structure of images, the Gestalt Principles are frequently used. Students and teachers may not always agree on the Gestalt Principles that an image contains, making assessment activities focusing on this topic very subjective. When working with large class sizes or when using multiple markers, this subjectivity becomes more problematic. A DBSCAN-based feature generation process and a multi-layer perceptron-based model are suggested in this study for categorising images as exhibiting the principle of proximity. When classifying images as containing aspects of proximity, the model displays a high level of accuracy. The model was integrated as a component of a larger process that alters the original input images to highlight areas that demonstrate proximity. The use of these annotated images can aid in giving students feedback. However, as the model was trained on a synthetic data set, its accuracy on real-world student submissions is yet to be tested.**

***Index Terms*—Machine learning, Educational technology, Computer graphics**

I. Introduction

The Gestalt Principles offer a way to direct students towards comprehending the organisation of images. However, marking student work in relation to the Gestalt Principles is very subjective, just like any design-related activity. In educational settings characterised by larger class sizes and the utilisation of multiple markers, the inherent subjectivity of the marking process is further intensified. This causes problems with class group marking consistency. Students who create images with the intention of adhering to the Gestalt Principles need direction on whether the images actually demonstrate the principle(s) they are attempting to illustrate.

The aim of this study is to devise a methodology capable of identifying and localising at least one Gestalt Principle within an image. The paper is organised as follows to support this objective. The Gestalt Principles are briefly discussed in Section II, along with examples of how they have been used in design-related coursework and how they might be assessed using machine learning techniques. The creation of the study's data set and how it was used to train a multi-layered perceptron (MLP)-based classification model are covered in Section III. A summary of the MLP's performance is given in Section IV, and a further discussion of the usefulness of such a model within a larger process is given in Section V. Section VI offers a conclusion to the study.

II. Related Work

A. The Gestalt principles

Three German psychologists set out on a quest to gain a better understanding of how people perceive the world [2]. They discovered a set of rules known as the Gestalt Principles, which makes an attempt to explain how the human brain processes visual information and creates order from the perceived disorder to help make sense of the perceived chaos.

The German word gestalt can be translated to English as *shape* [3, p. 239]. The phrase *to perceive the whole of a shape* has come to be synonymous with *to perceive the whole as more than the sum of its parts*. Thus, the Gestalt Principles (also known as Gestalt Laws) aid in describing how a person organises similar elements into groups within their field of vision, simplifies complex images, spots patterns, and combines disparate elements to create a whole. The commonly accepted Gestalt principles are similarity, continuation, closer, proximity, common region, figure/ground, symmetry, and common fate [4], [5]. Each of the Gestalt principles is illustrated in Figure 1.

The idea that similar elements frequently cluster together, regardless of how close they are to one another, is known as the principle of similarity. This classification can be determined based on the attributes of colour, shape, size, texture, or a combination thereof. None of the aforementioned attributes are required by the principle of continuity to classify elements. Instead, the components are categorised into groups based on whether they exhibit a seamless flow or pathway. The termination of the grouping occurs when any deviation from the established path is perceived as its conclusion. The proximity principle is a related idea. The principle of proximity simply states that visual elements that are close to one another or that have close proximity belong to the same group and does not rely on a perceived flow. The idea of a common region is connected to the idea of proximity. This principle emphasises that even though elements may be close to one another, the establishment of regional boundaries may prevent them from being grouped together. Figure 2 shows how the concept of a common region can take precedence over proximity-based natural groupings. The idea of common fate is another related principle that says objects moving in the same direction can be seen as a group regardless of their size, shape, or distance.

The human perceptual system has a tendency to fill in missing information in visual stimuli in order to perceive a coherent whole. The aforementioned phenomenon can be explained by the application of the Gestalt Principle of closure. Another closely interconnected principle is the concept of figure/ground. According to this principle, it is proposed that individuals possess a proclivity to perceive the foreground and background as discrete visual elements, thereby partitioning a scene into a figure and its corresponding background [6]. Both the principles of closure and figure/ground rely on the

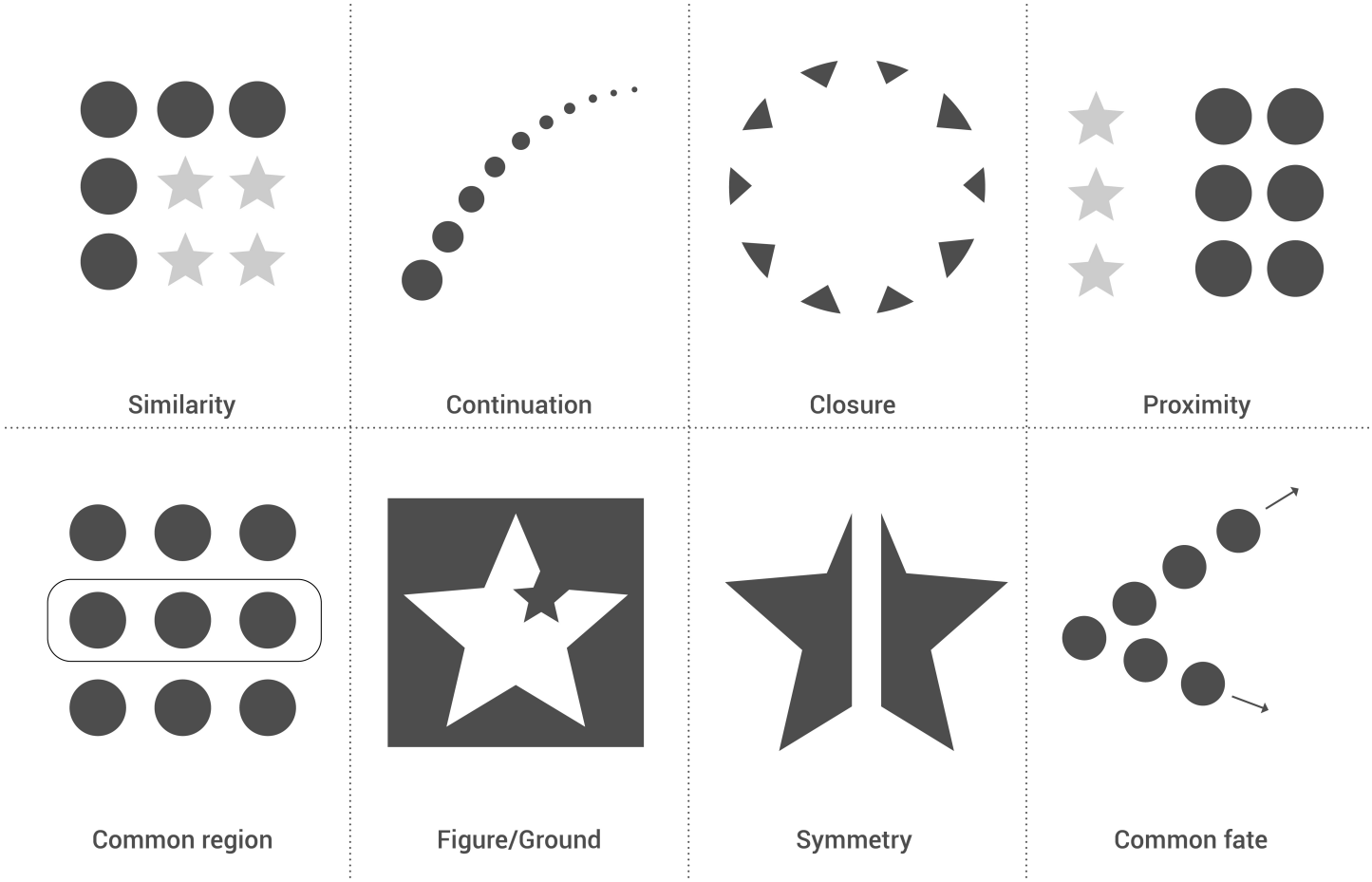


Fig. 1: Visualisation of the Gestalt Principles [1]

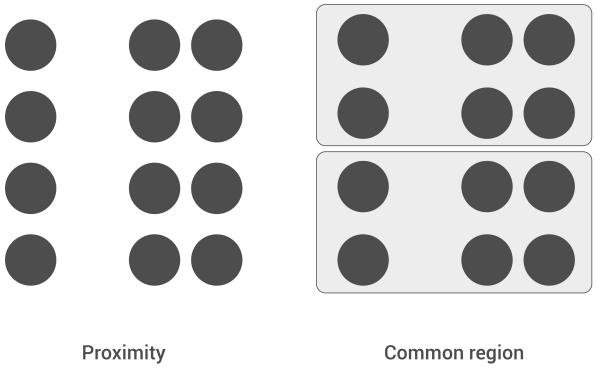


Fig. 2: Comparison of the Principles of Proximity and Common Region [1]



Fig. 3: Visualisation of the Principle of Proximity

cognitive processes associated with perceiving and interpreting negative space in visual elements. Negative space, in the context of visual arts, refers to the spatial region that encompasses the subject, object, or area of interest within an image. In contrast, positive space pertains to the subject, object, or specific area of focus.

The primary objective of this study is to examine the concept of proximity, which is considered one of the more straightforward principles within the Gestalt theory. Additionally, this principle is well-suited for applying fundamental machine learning techniques, such as clustering, to effectively group data points. This observation underscores the tendency of human beings to perceive a sense of association or connection between various elements when they are grouped together. The perception of proximity in an image is typically straightforward, as our visual system naturally organises elements that are in close proximity to one another. Figure 3 visualises the Gestalt principle of proximity.

B. Marking of designs

In the realms of information technology, computer science, and information systems, there are various sub-disciplines that focus on distinct aspects of the software development process. One particular sub-discipline within the field of design is user interface design (UID). User Interface Design (UID) is a discipline that centres around comprehending

user requirements and devising components that facilitate user comprehension and seamless access to said components. The primary objective of User Interface Design (UID) is to provide users with the necessary tools and resources to successfully accomplish their intended interactive tasks. The field of User Interface Design (UID) encompasses a range of considerations pertaining to the design of computerised systems, encompassing graphical interfaces as well as alternative modalities like voice control [7].

In the field of User Interface Design (UID), the Gestalt Principles play a crucial role in enabling designers to understand and predict user interpretations. Neglecting these principles tends to result in designs that display an unprofessional aesthetic and possess the capacity to convey unclear messages to the intended user(s) of a system. A comprehensive understanding of these principles is essential for students studying design, as they provide insight into the cognitive mechanisms through which individuals categorise and arrange elements, as well as identify patterns. Designers utilise these principles in order to establish a cohesive organisation of content on user interfaces and other design materials, with the objective of presenting a visually pleasing design that is readily understandable.

Within the framework of a UID academic programme or course, students may be assigned the responsibility of developing an interface or logo that adeptly incorporates one or more of the foundational principles of Gestalt. Upon the completion of the submission, the instructor(s) responsible for the specific subject matter are obligated to thoroughly analyse each individual submission and provide written feedback to the respective student.

A frequent source of disagreement between educators and learners pertains to the inherent subjectivity involved in the evaluation of these assignments. Given that design is fundamentally a subjective process [8], it is not unexpected that the evaluation of designs is also subjective, despite its theoretical basis. In numerous instances, it is possible for students to hold differing perspectives from their teachers regarding the incorporation (or omission) of Gestalt principles in a design. This issue may be exacerbated in an educational setting where multiple teachers are assigned to a student group, each of whom may hold their own perspective on a particular design.

To streamline the evaluation process, students are requested to clearly label their designs to highlight the integration of (a) specific Gestalt principle(s). This serves to alleviate a portion of the discord pertaining to the essence of the design, particularly in designs that exhibit multiple principles. The assigned assessor will proceed to analyse the image to determine if the design adheres to the principles outlined by the student. Following that, the instructor compiles feedback that encompasses an evaluation of the degree to which the design adheres to the specified principle(s), potential manifestations of additional principles, and recommendations for improving the design.

C. Machine learning

Machine learning (ML) is a subfield of artificial intelligence (AI) that involves the iterative training of models using input features. The purpose of this training is to enhance the models' ability to achieve a desired outcome, such as accurately classifying images based on their content. Machine learning is utilised in a diverse array of academic disciplines spanning multiple domains. This section focuses on the various applications machine learning is used for within the broader domains of image design, manipulation and education.

FontMatcher is a system that aims to provide recommendations and explanations regarding the optimal font pairing for a given design [9]. The application of machine learning has been employed for the automated generation of posters, serving as promotional material [10]. The developed system has the capability to integrate the preferences introduced through human intervention in order to update the designs of the posters. In another study, a model was proposed that can effectively estimate human attention on various types of images, including both designed and natural ones [11]. The model was integrated into a design application that had the capability to automatically adapt vector images to adhere to design constraints specified by the user. In recent times, there has been a growing emphasis on the development of fully automated techniques for generating images solely based on textual descriptions. Several companies have created systems that are built upon this principle [12], [13].

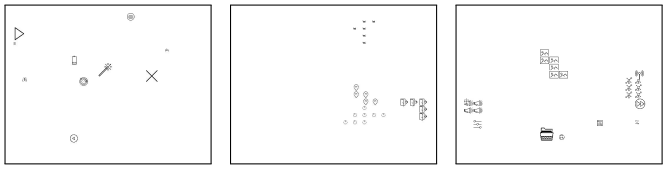
The application of machine learning techniques in conjunction with the Gestalt laws has been explored by proposing a methodology that involves generating Gestalt Regions of Interest (GROIs) to be used as input for a convolutional neural network (CNN) [14]. Convolutional neural networks (CNNs) have also been employed to classify the correctness of Kanzisa triangles (a type of optical illusion) by employing the Gestalt principle of closure [15]. Another study used a multi-layered perceptron-based network to identify whether an image, or a subset of an image, exhibits the Gestalt principle of symmetry [1]. A transformer-based framework has also been used to detect dashed curves within images [16]. The primary objective of this investigation was to develop a computerised system capable of recognising the Gestalt law of continuity.

III. A process for detecting proximity

To develop an automated method for identifying and evaluating proximity aspects in a design, the decision was made to utilise a clustering algorithm to generate input features for a multi-layered perceptron-based model. Clustering algorithms offer the capability to identify and delineate groupings within a given input data set. In order to evaluate the effectiveness of clustering in generating features for detecting the Gestalt principle of proximity, it was imperative to create test data and develop a methodology for extracting features from the test data. The subsequent section delineates the procedures entailed in the generation of test data and the extraction of features, which are subsequently utilised as input for a clustering algorithm.

A. Data set generation

To develop a model capable of discerning the presence of content demonstrating the principle of proximity in an image, it was imperative to acquire a dataset of images. In order to achieve this objective, a synthetic image dataset was generated by employing a blend of randomly generated geometric shapes and images sourced from a freely available icon pack. The purpose of creating the images was to present various examples, including those that do not inherently



(a) No-proximity (b) All-proximity (c) Some-proximity

Fig. 4: No-proximity, all-proximity and some-proximity images

exhibit proximity, those that exclusively showcase elements demonstrating proximity, and those that feature a combination of elements with and without proximity. Throughout the duration of the study, the aforementioned three categories of images shall be denoted as no-proximity, all-proximity, and some-proximity.

The process generated a quantity of 1000 images for each of the no-proximity, all-proximity, and some-proximity categories, resulting in a cumulative total of 3000 images. Each of the images was intentionally designed to possess a monochromatic colour scheme, specifically black and white, and adhere to a standardised resolution of 640x480 pixels. Examples of all the images are shown in Figure 4.

The image processing procedure entailed the generation of a set of features, which were subsequently utilised as input for the clustering algorithm. A preprocessing procedure was conducted on each image in order to generate features. This procedure involved converting the images to grayscale and subsequently applying binarization. The initial images were initially presented in a black and white format. However, the process of binarization is utilised to ensure that the images solely consist of values that distinctly represent either black or white. The pre-processing stage encompasses a step in which all images are resized to dimensions of 640 x 480. Nevertheless, it is important to acknowledge that the resizing process was not required for the images produced for the synthetic dataset employed in this study.

The process of binarization involves the application of a threshold to all RGB colour values, specifically at a median byte value of 127. Values that are less than 127 are assigned a value of 0, while values that are equal to or greater than the threshold are assigned a value of 255. Additionally, this is carried out to ensure the feasibility of applying the feature generation process to colour images, which could potentially be submitted by students if the model is intended for use in a real-world context.

Each binary image is then processed using OpenCV's findContours() method to identify the outlines of objects in the input image and calculate their centroid coordinates [17]. Although it is possible to filter the identified shapes according to their size, this step was not taken to ensure that the details of all detected shapes are available to the clustering algorithm. The principle of proximity is also not concerned with the size and shape of the individual elements, but only whether they are located close to one another (in location) in the image. For every image, a CSV file is generated, with each row representing the centroid x-and-y coordinates of each individual shape detected in the image.

B. Training the Model

The Density-Based Spatial Clustering of Applications with Noise (DBSCAN) algorithm is widely employed in the field of data clustering to partition datasets into distinct clusters by leveraging density characteristics. In contrast to the k-means algorithm, DBSCAN does not necessitate the specification of the number of clusters and exhibits the ability to identify clusters of diverse shapes and sizes. This method demonstrates a high level of efficacy in managing data sets that contain irregularly shaped clusters and noise. As the number of proximity groupings in an image is not known before the image is analysed, DBSCAN was selected as the clustering algorithm used in this study to generate basic features related to proximity.

Using the CSVs files associated with each of the 3000 training images, the DBSCAN clustering algorithm was applied to each set of input features using Python and the Scikit Learn (*sklearn*) implementation of the DBSCAN algorithm [18]. The DBSCAN algorithm's arguments can be configured to determine how close points should be (with regards to their enclosing area) to be considered part of the same cluster and how many samples should be within this area in order for the data points not to be considered noise. Although DBSCAN is able to automatically generate clusters without requiring a specific required number of clusters to be defined, it provides no measure as to the quality of the generated clusters. For that purpose, *sklearn* also provides the ability to calculate a Silhouette Coefficient (silhouette_score), which serves as a measure of the quality of clustering results. This score ranges from -1 to 1, with values closer to 1 indicating that a data point is associated with the correct cluster and a value closer to -1 indicating that it is not.

For each processed image (CSV), the number of clusters identified, as well as the average Silhouette Coefficient of the clusters are calculated and stored, along with a label of *No* or *Yes*. Using scaling, the number of clusters feature and the Silhouette Coefficient are both scaled to a range of 0 to 1. Each CSV file associated with a no-proximity image received the label of *No*, whereas all other images were labelled as *Yes*. This process yielded a data set containing 3000 individual entries, of which 1000 are labelled as *No* and 2000 as *Yes*.

The dataset was employed to create a training dataset as well as a test dataset. During the training phase of the experiment, 90% of the available data set was utilised to train a classification model using a multi-layer perceptron (MLP) architecture. Following this, the remaining 10% of the dataset was utilised to assess the accuracy of the trained model. Neural networks consist of distinct elements referred to as nodes or perceptrons. The multi-layer perception is a neural network type that is created as a computational model consisting of multiple layers of interconnected perceptrons. It is characterised by having at least one input layer and one output layer [19, p. 237].

The construction of the MLP involved employing a methodology that incorporated several techniques. These techniques included the utilisation of 10-fold cross-validation, adaptive learning methods, a maximum of 3000 iterations for each training process, and a termination criterion of 50 iterations in cases where no noticeable improvements were observed in the model. The technique of cross-validation is extensively employed in the domain of machine learning. The process entails dividing a provided dataset into numerous folds or subsets. The model is subsequently trained on a majority of these subsets, typically excluding one, and subsequently evaluated on the remaining subset. The cross-

validation process entails iteratively traversing the dataset for a number of times equivalent to the number of folds. This practice guarantees the inclusion of all data points present in the dataset. The calculation of the overall model error can be obtained by taking the average across the different folds [20, p. 46]. The model was developed and evaluated in Python using the *sklearn* library's *MLPClassifier* class and the cross_validate function.

IV. Evaluation

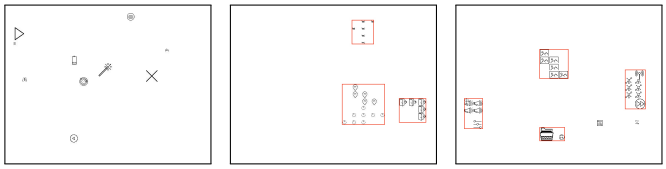
The MLP model employed in this study was subjected to testing using a subset that accounted for 10% of the entire synthetic dataset. The model exhibited a notable degree of accuracy in categorising images as containing examples of proximity or not; achieving a success rate that consistently surpassed 90% across multiple experimental iterations. This implies that the task of identifying elements representing proximity is not inherently difficult, and it is feasible for an automated procedure to replace a human in carrying out this task. To validate this assertion, it was necessary to ascertain the specific levels at which individuals could reach consensus regarding the subjective endeavour of discerning proximity.

To perform a thorough assessment, a sample size of 100 images was selected at random from the collection of test images, which included all three types of images. The authors of this paper conducted a comprehensive analysis of the 100 images to evaluate the extent to which each image exhibited the Gestalt principle of proximity, relying on their subjective assessment. The authors did not possess any prior knowledge regarding the selection of images, as the images were chosen in a random manner. Furthermore, the authors lacked knowledge regarding the quantity of images chosen to illustrate the concepts of no-proximity, all-proximity, or some-proximity. The analyses were conducted by the authors in a manner that maintained independence from each other. The authors acknowledge that this approach incorporates a level of subjectivity, but it replicates an authentic evaluation context in which student designs would also be evaluated based on predetermined conceptions or guidelines. After the authors finished their classifications, they proceeded to compare them both internally and with the classifications generated by the model. This comparison involved scrutinising which classifications the observers (two humans and the trained model) were in agreement on (or not), as to whether the example image exhibited the principle of proximity.

The results obtained from this experiment provide evidence that human observers possess a significant ability to perceive proximity in visual stimuli, as indicated by the substantial consensus among the participants. The human participants reached a consensus on 94% of the images regarding their demonstration of the principle of proximity. Upon individual analysis, it was observed that both human observers exhibited concurrence with the model on over 90% of the images. This finding implies that the model's ability to identify proximity is comparable to that of human individuals performing the identical task.

V. Discussion

Based on the findings obtained from the model evaluation, it appears that the task of classifying images as either containing proximity or not is relatively straightforward when the entire image is taken into account. The subjectivity of the image increases when individual elements are examined in isolation. The model-based approach may possess an advantage in this particular activity due to its ability to meticulously examine smaller components of the image at a more detailed level. This enables the system to more effectively detect and recognise proximity aspects compared to human observers.



(a) No-proximity (b) All-proximity (c) Some-proximity

Fig. 5: Detecting Proximity

Utilising an automated process for feedback facilitates the identification of areas of proximity in a student's work. This process has the capability to automatically highlight any regions in the image that demonstrate proximity. As the process for generating an image's features contains the locations of individual blobs, it's possible to use this information, along with the cluster details to demonstrate individual areas of proximity within an image, as depicted in Figure 5. Educators, such as teachers or lecturers, can utilise a process (along with a corresponding tool) of this nature to produce feedback visuals for students prior to composing their own written feedback, thereby enhancing the effectiveness of the generated image. By explicitly delineating the specific region that demonstrates close proximity, the provision of feedback would become more focused and potentially mitigate the disputes arising from the subjective nature of evaluating design elements. In an alternative scenario, the tool could be made available to students as a means of assisting them in assessing the proximity elements within their own work prior to submitting an image for evaluation.

VI. Conclusion and Future Work

The primary objective of this study was to develop a methodology capable of identifying and analysing at least one Gestalt Principle that is evident in an image. Furthermore, the objective of the study was to offer feedback regarding the specific location at which the identified principle was observed. The stated objective was effectively achieved by employing a multi-layer perceptron model that was trained using features derived from the application of DBSCAN clustering to a synthetic data set. Following this, the model was incorporated into a more comprehensive framework, enhancing its ability to identify areas of closeness within images and also offering annotations to highlight regions that demonstrate the concept of proximity.

The model demonstrated a notable degree of accuracy in categorising images according to their conformity to the Gestalt principle of proximity. The degree of agreement between human participants and the model constructed for this investigation is also very high. The automated nature of this process makes it particularly suitable for identifying areas of proximity, either as an alternative to or in conjunction with a human evaluator. The procedure has the capacity to attain a level of accuracy similar to that of a human observer. Furthermore, it is capable of consistently executing this task without succumbing to fatigue. This approach could offer

significant benefits in educational environments characterised by large class sizes and multiple evaluators, as it has the potential to reduce the inherent subjectivity that arises from individual human evaluators.

The second component of the study's objective requires the provision of feedback regarding the precise location within the image where the Gestalt principle was detected. The issue was resolved by modifying the input images to emphasise the areas that demonstrate close proximity. The feedback method presented herein provides a coherent and targeted approach, facilitating students' ability to readily discern the precise regions within their visual representation that exhibit characteristics of proximity.

Future research endeavours will entail the application of the process and machine learning model devised in this study within an educational context. The evaluation will examine the effectiveness of the process in assisting evaluators in the evaluation of student submissions, as well as its capacity to aid individual students in their respective design processes. After evaluating the efficacy of the model, it will be integrated into a unified system capable of detecting all the Gestalt Principles that are present in a given image.

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